Abdelrahman Khalifa

Embedded Linux Engineer | Linux, Yocto, C, Rust, Open Source

abdelrahman.khalifa.gad@gmail.com | +20 114 250 5042 | Cairo, Egypt

linkedin.com/in/abd002 | github.com/Abd002 | codeforces.com/profile/_Khalifa

SUMMARY

Embedded software engineer who works upstream. Built custom Linux images with Yocto and Buildroot, wrote a Linux kernel module for ARM, and booted ARM boards in QEMU with U-Boot, device trees and BusyBox root file systems. Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting), with 11 PRs merged upstream in C and Rust, plus merged work in Zephyr RTOS, Boa and Rust Clippy. Owns the Zephyr firmware and firmware-update path for EKSON's motion simulators. Used to remote, review-driven work.

EXPERIENCE

Google Summer of Code Contributor | **The Linux Foundation (OpenPrinting)** | **May 2026 – Oct 2026**

Open-source project: Printers page for COSMIC Settings (Linux desktop), in Rust; passed the final evaluation | github.com/Abd002/GSOC-2026

- Built the **Printers page for COSMIC Settings** and a standalone app for the Linux printing stack (CUPS, IPP, DNS-SD/Avahi, Printer Applications), worked remotely with mentors from OpenPrinting and System76 (cosmic-settings #2182, in review).
- Ran the C FFI backend as a separate varlink IPC service in `cosmic-settings-daemon`, so a crash in native code cannot take down the Settings app.
- Got **11 PRs merged upstream** (cups-rs, #164, #165): ported cups-rs to libcups3 with a GitHub Actions matrix across architectures and CUPS versions, added DNS-SD FFI bindings, and fixed crashes and a leaked client in libcups' Avahi code (C).

Embedded Software Engineer | **EKSON** | **Aug 2025 – Present**

Software for 3-DoF and 6-DoF pneumatic motion simulators: firmware, desktop app, backend and CI

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS for devicetree-based drivers and easier porting; PID valve control with IMU feedback over I2C, safety stops and a watchdog.
- Built remote operation and firmware updates: the desktop app polls a Node.js/MongoDB backend over REST and flashes new firmware with esptool; GitHub Actions CI. Cut the app's CPU use and latency by about 40% (measured).

Embedded Linux Systems Intern | **Orange Digital Center (ODC)** | **Dec 2024 – Jan 2025**

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing. Scored 100% in the final evaluation.

OPEN SOURCE PROJECTS

- **Zephyr RTOS** (Mar 2025): implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules, reviewed and merged (issue #86977).
- **Rust ecosystem** (Dec 2025): Boa JavaScript engine string optimization (PR #4553) and a Rust Clippy lint fix (PR #16214), both merged.

PROJECTS

- **Kernel Monitor** (C, Linux kernel, ARM): character device driver that walks the process list under RCU and exports per-task IDs, memory, threads, context switches and CPU time; cross-compiled for ARM and run on QEMU vexpress-a9 with a custom kernel, device tree and BusyBox ext4 root file system. GitHub
- **A/B firmware updates** (graduation project; STM32, ESP32): an ESP32 gateway sends the image over UART with UDS services; the STM32 bootloader verifies it with the hardware CRC unit and uses two banks with rollback. GitHub
- **IoT Device Communication** (C++, Linux, QEMU): TCP client–server classes; ARM clients run on two QEMU boards booted with U-Boot over TFTP and a bridged TAP network. GitHub

TECHNICAL SKILLS

Embedded Linux: Yocto, Buildroot, U-Boot, Linux kernel modules and drivers, device trees, BusyBox root file systems, QEMU (ARM), cross-compilation, Raspberry Pi

Languages: C, Rust, Modern C++, Python, Bash

Firmware: Zephyr (devicetree, west), FreeRTOS, ESP-IDF, ARM Cortex-M (STM32, TM4C), ESP32-S3, bootloaders, firmware updates

Linux & tools: Debian/Ubuntu, Arch (CachyOS), CUPS/IPP, DNS-SD/Avahi, varlink/D-Bus IPC; Git, GitHub Actions CI, CMake, Make, Cargo, upstream code review

EDUCATION

Bachelor of Science (B.Sc.), Computer and Systems Engineering | **Minia University** | **2020 – 2025**

- Grade: Very Good with Honors. Codeforces Expert; 4-time ECPC finalist (2021–2024).